PREACTION/DELUGE SPRINKLER SYSTEMS INSPECTION, TESTING, AND MAINTENANCE

Property Name: **Inspector:**

Property Address: **Contract No.:**

Property Phone Number: **Date:**

Inspection Frequency: **Daily** **Weekly** **Monthly** **Quarterly** **Annual** **Five-Year**

Inspections: Daily			**Preaction and Deluge Valve (Cold Weather/Heating Season Only)**
Yes	No	N/A	Enclosure, not equipped with a low temperature alarm, is inspected during cold weather to verify a minimum temperature of 40°F (4°C)
Inspections: Weekly			
			Backflow
Yes	No	N/A	Isolation valves — open position and locked or supervised
Yes	No	N/A	RPA and RPDA — differential-sensing relief valve operating correctly
			Control Valves
Yes	No	N/A	In the correct (open or closed) position
Yes	No	N/A	Sealed
Yes	No	N/A	Accessible
Yes	No	N/A	Post Indicator Valves (PIVs) are provided with correct wrenches
Yes	No	N/A	Free from damage or leaks
Yes	No	N/A	Proper signage
			Preaction and Deluge Valve
Yes	No	N/A	Enclosure, where equipped with low temperature alarm, is inspected during cold weather to verify a minimum temperature of 40°F (4°C)
			Master Pressure-Regulating Device
Yes	No	N/A	Downstream pressures are in accordance with design criteria psi
Yes	No	N/A	Supply pressure is in accordance with design criteria psi
Yes	No	N/A	Free of damage or leaks
Yes	No	N/A	Trim in good operating condition
Inspections: Monthly			
Yes	No	N/A	Gauges are operable and not physically damaged
Yes	No	N/A	Gauges — normal air or nitrogen pressure maintained (not supervised)
Yes	No	N/A	Gauge on system side of dry valve reads proper ratio of air or nitrogen (not supervised) psi
Yes	No	N/A	Gauge on quick-opening device reads the same as system side dry valve gauge (not supervised) psi

PREACTION/DELUGE SPRINKLER SYSTEMS INSPECTION, TESTING, AND MAINTENANCE (Continued)

Inspections: Monthly			**Control Valves (Locked or Supervised)**
Yes	No	N/A	In the correct (open or closed) position
Yes	No	N/A	Locked or supervised
Yes	No	N/A	Accessible
Yes	No	N/A	Post Indicator Valves (PIVs) are provided with correct wrenches
Yes	No	N/A	Free from damage or leaks
Yes	No	N/A	Proper signage
			Preaction/Deluge Valve
Yes	No	N/A	Free from physical damage or leaks
Yes	No	N/A	Electrical components are in service
Yes	No	N/A	Trim valves are in the correct (open or closed) position
Yes	No	N/A	Valve seat is not leaking
Inspections: Quarterly			
Yes	No	N/A	Gauges — normal air or nitrogen pressure maintained when supervised at a constantly attended location psi
Yes	No	N/A	Gauge on system side of dry valve reads proper ratio of air or nitrogen when supervised at a constantly attended location psi
Yes	No	N/A	Gauge on quick-opening device reads the same as system side dry valve gauge when supervised at a constantly attended location psi
Yes	No	N/A	Gauge on supply side of valve reads normal psi
Yes	No	N/A	Waterflow alarm and supervisory devices are free of damage
			Fire Department Connections
Yes	No	N/A	Visible and accessible
Yes	No	N/A	Coupling/swivels operate correctly
Yes	No	N/A	Plugs/caps are in place
Yes	No	N/A	Gaskets are not damaged
Yes	No	N/A	Identification signs are in place
Yes	No	N/A	Check valve is not leaking
Yes	No	N/A	Automatic drain valve in place and operating correctly
Yes	No	N/A	Clapper operates correctly
Yes	No	N/A	Interior is clear of obstructions (unless locked)
Yes	No	N/A	Visible piping supplying the fire department connection is undamaged
			Pressure-Reducing Valve
Yes	No	N/A	In the open position and not leaking
Yes	No	N/A	Maintaining downstream pressure
Yes	No	N/A	In good condition, with handwheel installed and unbroken

PREACTION/DELUGE SPRINKLER SYSTEMS INSPECTION, TESTING, AND MAINTENANCE (Continued)

Inspections: Quarterly			**Control Valves (Electronically Supervised)**
Yes	No	N/A	In the correct (open or closed) position
Yes	No	N/A	Electronically supervised
Yes	No	N/A	Accessible
Yes	No	N/A	Post Indicator Valves (PIVs) are provided with correct wrenches
Yes	No	N/A	Free from damage or leaks
Yes	No	N/A	Proper signage
Inspections: Annual			
Yes	No	N/A	Hydraulic design information sign is securely attached to riser and legible
			Sprinklers (visible)
Yes	No	N/A	No damage or leaks
Yes	No	N/A	Free of corrosion, foreign material, or paint
Yes	No	N/A	Installed in proper orientation
Yes	No	N/A	Fluid in glass bulbs
Yes	No	N/A	Spare sprinklers — proper number and type, including installation wrench
Yes	No	N/A	Loading — sprinklers are free of dust
Yes	No	N/A	No paint or coating other than that applied by the manufacturer
Yes	No	N/A	Escutcheons/cover plates are present and installed correctly
Yes	No	N/A	Minimum clearance between sprinklers and storage
			Hangers/Seismic Bracing
Yes	No	N/A	Not damaged or loose
			Pipes and Fittings
Yes	No	N/A	In good condition and no external corrosion
Yes	No	N/A	No leaks or mechanical damage
Yes	No	N/A	Correct alignment and no external loads
			Fire Department Connections
Yes	No	N/A	Interior of connection with locked plugs or caps is free of obstructions
			Preaction/Deluge Valve
Yes	No	N/A	Interior inspection following trip test
Yes	No	N/A	Detection device condition inspection
			Building
Yes	No	N/A	Prior to onset of freezing weather, all openings are closed and water-filled pipe is not exposed to freezing temperatures
Yes	No	N/A	Heat trace is per manufacturer's instructions
Yes	No	N/A	Low temp alarm is free of physical damage

PREACTION/DELUGE SPRINKLER SYSTEMS INSPECTION, TESTING, AND MAINTENANCE (Continued)

Inspections: Five Years

Yes	No	N/A	Obstruction inspection — no foreign or obstructing material is found
Yes	No	N/A	Check valve — internal moves freely and in good condition
Yes	No	N/A	Internal inspection of preaction/deluge valve strainers, filters, restricted orifices, and diaphragm chambers
Yes	No	N/A	Internal inspection of valves that can be reset without removal of faceplate
Yes	No	N/A	Internal inspection of backflow

Test: Quarterly

Yes	No	N/A	Alarm devices — water motor gong
Yes	No	N/A	Detection system low air pressure supervisory device
Yes	No	N/A	Main drain test, if the sole supply is through a backflow preventer or pressure-reducing valve
			Static psi Residual psi
Yes	No	N/A	Do results differ by more than 10% from previous test?
Yes	No	N/A	Priming water — test level
Yes	No	N/A	Low air alarm — test per manufacturer's instructions
			Master Pressure-Regulating Device
Yes	No	N/A	Partial flow test performed to exercise valve

Test: Semiannual

Yes	No	N/A	Valve supervisory switch(es) function
Yes	No	N/A	Alarm devices — inspector's test or bypass opened and observed waterflow

Test: Annual

Yes	No	N/A	Valve supervisory switch(es) function
Yes	No	N/A	Low temperature alarm (if installed) at the beginning of heating season
Yes	No	N/A	All control valves operated through full range of motion and returned to normal position
Yes	No	N/A	Pressure reducing valve partial flow test
Yes	No	N/A	Automatic air maintenance device functional
Yes	No	N/A	Valve status test performed
Yes	No	N/A	Backflow preventer — forward flow test at a minimum flow rate of the system demand
			Main Drain Test
			Static psi Residual psi
Yes	No	N/A	Do results differ by more than 10% from previous test?

PREACTION/DELUGE SPRINKLER SYSTEMS INSPECTION, TESTING, AND MAINTENANCE (Continued)

Test: Annual			**Full Flow Trip Test (Deluge Valve)**
Yes	No	N/A	Unobstructed discharge from all nozzles
			Pressure reading at deluge valve psi
Yes	No	N/A	Compare if pressure readings to hydraulic design/water supply meets requirements
Yes	No	N/A	Manual release functions correctly
Yes	No	N/A	Valve status test performed
Yes	No	N/A	Pressure reading at most remote nozzle or sprinkler psi
Yes	No	N/A	Air maintenance device functions correctly (if provided)
			Preaction Partial Flow Trip Test
Yes	No	N/A	Preaction valve — trip test with partially open control valve
			Water pressure (psi) Air pressure (psi)
			Tripping air pressure (psi) Trip time (sec)
			Water delivery time (min.)
Yes	No	N/A	Results compared to previous results
Test: Three Year			**Preaction Full Flow Trip Test**
			Preaction valve — trip test with open control valve
			Water pressure (psi) Air pressure (psi)
			Tripping air pressure (psi) Trip time (sec)
			Water delivery time (min.)
Yes	No	N/A	Results compared to previous results
Yes	No	N/A	Preaction system tested for air leakage
Test: Five Years			
Yes	No	N/A	Gauges tested or replaced
Yes	No	N/A	Sprinkler pressure-reducing valve — flow test and comparable to previous results
			Fire Department Connections
Yes	No	N/A	Piping from fire department connection to fire department connection check valve has been hydrostatically tested at 150 psi (10 bar) for at least 2 hours
Routine Maintenance			
Yes	No	N/A	Sprinklers/pilot sprinklers tested or replaced per appropriate testing schedule
Yes	No	N/A	OS&Y — stems lubricated annually
Yes	No	N/A	Leaks causing drops in supervisory air pressure or electrical malfunctions causing alarms fixed
Yes	No	N/A	Interior of valve cleaned after trip test and internal inspection
Yes	No	N/A	Operate axillary drains after system operation and before freezing conditions

PREACTION/DELUGE SPRINKLER SYSTEMS INSPECTION, TESTING, AND MAINTENANCE (Continued)

Comments

Signature: Date:

Contractor Name:

Contractor Address:

License/Certification No.:

PREACTION/DELUGE SPRINKLER SYSTEMS INSPECTION, TESTING, AND MAINTENANCE (Continued)

This form covers a 6-month period.

Year: **System:**

Location:

General

1. If valves are sealed, note "yes" in this block. If any are not sealed, reseal and note "resealed" in this block.
2. Gauges for dry, preaction, and deluge systems must be inspected for normal air and water pressures.

3–6. Record pressure readings in psi (bar). A loss of more than 10% should be investigated.

7. Record any notes about the system that the inspector believes to be significant. Place a number in the box and corresponding note in space provided below.

Date	Inspector	Valves Sealed (1)	Gauges (2)	Alarm Valve OK (3)	Dry Pipe Valve (4)		Preaction Valve (5)		Deluge Valve Water Pressure (6)	Notes (7)
					Air Static Press.	Water Static Press.	Air Static Press.	Water Static Press.		

Notes: